SESSION 5 REFERENCE GUIDE: DATA CLEANING, MISSING VALUES & TYPE CASTING

This guide provides a reference for detecting missing values, imputing numeric entries, removing duplicate rows, standardizing strings, and parsing dates in Pandas.

1. Detecting & Handling Missing Values (`NaN`)

Pandas represents missing data using `NaN` (Not a Number).

```python
import pandas as pd
import numpy as np

# Count missing values per column
df.isnull().sum()

# Drop rows where ANY column has a missing value
df_clean = df.dropna()

# Drop rows where a SPECIFIC column has missing values
df_clean = df.dropna(subset=['critical_column'])

# Impute missing numeric values with a fixed scalar or calculated metric
median_val = df['numeric_col'].median()
df['numeric_col'] = df['numeric_col'].fillna(median_val)
```

2. Removing Duplicate Records

System logging retries often create duplicate rows.

```Python
# Count exact duplicate rows
duplicate_count = df.duplicated().sum()

# Drop exact duplicate rows and reset row index
df = df.drop_duplicates().reset_index(drop=True)

# Drop duplicates based on specific key columns
df = df.drop_duplicates(subset=['order_id', 'customer_id'])
```

3. String Cleaning via `.str` Accessor

Standardize text categories before performing grouping or filtering.

Method	Purpose	Example Code	Output Example
`.str.strip()`	Removes leading/trailing whitespace	`df['cat'].str.strip()`	" apparel " -> "apparel"
`.str.lower()`	Converts string to lowercase	`df['cat'].str.lower()`	"Apparel" -> "apparel"
`.str.upper()`	Converts string to uppercase	`df['cat'].str.upper()`	"apparel" -> "APPAREL"
`.str.title()`	Capitalizes first letter of each word	`df['cat'].str.title()`	"electronics" -> "Electronics"

4. Date Parsing & Type Casting

Explicit data types ensure calculations work as expected.

```Python

# Convert text string column to explicit Datetime object
df['order_date'] = pd.to_datetime(df['order_date'])

# Extract date components from Datetime column
df['year'] = df['order_date'].dt.year
df['month'] = df['order_date'].dt.month

# Cast numeric column to float or integer
df['quantity'] = df['quantity'].astype(int)
df['price'] = df['price'].astype(float)
```

5. The Data Cleaning Workflow

```
[Inspect df.isnull()] --> [Drop Duplicates] --> [Clean String Formatting] --> [Parse Datetimes] --> [Impute Missing Values]
```